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RESEARCH ARTICLE

Deep Learning for Automatic Defect Detection in PV Modules Using Electroluminescence Images

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ABSTRACT Solar energy, in the form of photovoltaic (PV) panels, is important for achieving clean energy solutions. The photovoltaic health index must be monitored and improved because of the high demand for green energy. Unfortunately, defective solar cells are a significant source of performance degradation in photovoltaic (PV) systems. Experts often manually analyze electroluminescence (EL) images by visually inspecting them, which is personal, time-consuming, and requires extensive expertise. This work presents a comparative analysis of YOLOv8 and an Improved YOLOv5 for an automatic PV defect detection system in EL images in which Global Attention Module (GAM) is incorporated into the traditional YOLOv5s model for better object representation. Adaptive Feature space fusion (ASFF) was added to YOLOv5's original structure for feature fusion. The Distance Intersection over Union (Non-Maximum) Suppression (DIoU-NMS) is aggregated to produce a more accurate bounding box. The ELDDS1400C5 dataset was used to train and evaluate the proposed system. Experiments on the ELDDS1400C5 test set revealed that the Improved YOLOv5 algorithm achieved a mean Average Precision of 76.3% (mAP@0.5), which is a 2.5% improvement over the standard YOLOv5 algorithm for detecting faults in PV modules in EL images. Furthermore, the experimental results demonstrated that Test Time Augmentation (TTA) significantly increased the mAP@0.5 to 77.7%, surpassing the YOLOv8 model, which achieved 77.5% under the same conditions.

INDEX TERMS ELDDS1400C5 dataset, electroluminescence images, photovoltaic panels, test time augmentation, YOLOv5, YOLOv8.

I. INTRODUCTION

In its diverse types and forms, such as natural gas, oil, and coal, energy is an essential factor in economic development and must be considered. The continuously increasing consumption of natural gas, oil, and coal has resulted in the consumption of approximately half of the country's conventional energy reserves. As a result, most countries have recently turned to renewable energy, particularly solar energy, to compensate for the scarcity of electric grid [1]. Many countries now use solar energy as an alternative to conventional energy to reduce reliance on fossil fuels in transportation and power plants and provide electricity to isolated areas such as mountains and deserts. By the end of 2015, the solar cell industry had grown significantly compared to traditional energy sources, extending to more than 30 countries [2]. The total amount of solar electricity produced in 2017 was 178 GW. This is sufficient to satisfy 1% of the world electricity demands [3]. According to the International Energy Agency, PV systems will be the principal source of electricity worldwide by 2050 [4]. Maintaining the solar cell power efficiency is an essential factor in enabling a sustainable and economical solar system. During the manufacturing process, there are two methods for determining the quality of the solar panels: flash and EL testing. The flash test measures the output performance of a panel by exposing it to a short, intense flash of light with a power spectrum similar to that of the Sun. Subsequently, the resulting I-V curve was used to calculate the maximum possible performance of the panel. EL imaging is another technique used to identify the defects in PV modules. Photons are emitted when a solar panel is exposed to electricity during EL testing. These images were

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analyzed using infrared or near-infrared cameras to detect and characterize various defects. The EL images were obtained in the dark as a grayscale images. In EL images, microcracks and other defects are depicted as dark grey lines or regions. Visual inspection of EL images is costly and time-consuming. In addition to being limited to a small scale, visual inspection requires appropriate study and experience. Extending visual inspection to a larger scale requires automatic detection methods [5]. Wind, snow, thermo-mechanical stresses, and human traffic during regular operations and maintenance, fabrication, shipping, and installation processes can all lead to cracks in PV modules [6]. In addition, PV modules are prone to other defects such as ShortCircuitString, ShortCircuitCell/LowPowerCell, MicroCrack, DarkSpot, Scratch, and Contamination. One of the primary causes of power outages in PV modules is cell cracking [7]. As a result, PV modules and cells must be monitored during manufacturing and later stages to achieve maximum output [8]. As a result, PV module maintenance and inspection have received significant attention in recent years.

Owing to its simplicity and speed, the YOLO v5 detection model is widely used for the object detection field. Many researchers from various fields have improved the original YOLO v5 model to cope with their target detection. For example, Li et al. developed YOLO-FIRI, which is an improved YOLO v5 object detector for infrared images. They added the cross-stage-partial connections (CSP) module to the feature extraction network to optimize shallow features. Furthermore, they used an improved attention module in the residual block to focus more on the targets and inhibit the background. They also used multiscale detection to improve the detection of small objects by the model. The proposed model was assessed using the KAIST and FLIR datasets. The proposed model outperformed YOLOv4 and YOLO-FIR in terms of accuracy, speed, parameter reduction, and computational cost optimization [9]. YOLOv5-Aircraft, an improved YOLO v5-based aircraft target detection model, was proposed by Luo et al. [10]. They combined centering and scale calibration to improve the feature extraction ability of the model. They replaced the residual block in the backbone feature extraction network with a CSandGlass block to reduce information loss. They demonstrated that the proposed model significantly improved the speed and accuracy of aircraft detection in remote sensing images, even when the background was complex. TPH-YOLOv5 was proposed by Zhu et al. [11] to detect objects in drone-capture scenarios. They added another prediction head to detect objects of various sizes. Furthermore, the original prediction head is replaced with the transformer prediction head. They used the CBAM attention mechanism to deal with dense-object scenarios. According to the experimental results, the TPH-YOLOv5 detection performance improved by approximately 7% compared with the original YOLOv5 performance. Jin et al. [12]improved YOLOv5 for defect detection in the adhesive structures. They fine-tuned the loss function and structure of the DCGAN to generate high-quality defect images and solve the unbalanced dataset problem. They used the EIoU loss function to replace the original localization loss. They added the CBAM attention after the C3 block to improve the feature extraction ability of the model. Compared to the original YOLOv5, the proposed model improved precision by 10.6%. Wang et al. [13] proposed an enhanced YOLOv5 method for detecting Prohibited Items in Baggage Security. They used a transformer in the final convolutional block of the backbone to improve the feature extraction ability. Furthermore, to address the issue of complex backgrounds in X-ray baggage images, they included a global attention mechanism (GAM) module in the head branch. Finally, the adaptive spatial feature fusion algorithm was utilized in the prediction section to generate more accurate bounding boxes. The experimental results on the OPIXray dataset revealed that the speed of the proposed model is 47 frames per second to meet practical and real-time requirements. YOLOv5-SLL, an improved YOLOv5 model for detecting American sign language letters, was proposed by Li et al. [14]. They began by creating the Sign Language Letters (SLL) dataset, which included 3373 images of annotated sign language letters. They then added CBAM to the YOLOv5 network structure to reduce background noise and focus on hand features. Furthermore, they used the soft-Non Maximum Suppression (soft-NMS) algorithm to deal with the issue of overlapping hands. The proposed YOLOv5s-SLL model outperformed the original YOLOv5 model by 2.03% in (mAP). It also outperformed the performance of the YOLOv3 and SSD models. Dai et al. [15] improved the YOLOv5 model to detect regular and irregular tunnel cracks. They annotated tunnel cracks using a novel labeling method. They also used mosaic data augmentation techniques to improve the detection accuracy of the model. They compared different attention mechanisms, such as the Coordinate Attention (CA) module, Squeeze-and-Excitation (SE) module, and Efficient Channel Attention (ECA) module to the YOLOv5m model. Among the other attention mechanisms, the Integration of the CA module had the highest mAP of 79.7%. The experimental results show that the proposed YOLOv5m-CA model outperforms the Faster R-CNN and SSD models by 23.2% and 24%, respectively. Zhang et al. [16] developed an improved YOLOv5 model for electric bicycle detection for optimal management of charging and parking. Using the YOLOv5s as the starting point, they proposed a lighter network backbone based on GhostNet. They used CIOU loss to generate more accurate bounding boxes. They also used quality focal loss (QFocal Loss) to improve the detection accuracy. In addition to reducing model complexity and optimizing computational resources, the proposed model achieved an mAP of 93.33%. In another study, Cao et al. [17] integrated the GhostNet module into YOLOv5 neck for a light tea bud detection model. They also included a CA attention mechanism for the best feature extraction. Furthermore, a bi-directional feature pyramid network (BiFPN) was used in the feature fusion neck layer. Finally, they replaced the default localization loss with the efficient intersection over union

(EIOU) loss to improve detection accuracy. As a result, the proposed GhostNet-YOLOv5 outperformed Faster R-CNN, YOLOv5, and YOLOv5-Lite with an mAP of 76.31%.

The following are the main contributions of this paper:

1. The Global Attention Mechanism (GAM) was incorporated into YOLOv5s to improve the network's overall feature extraction ability.

2. Replacement of the default NMS with Distance Intersection over Union (DIoU), which considers the overlap area and the center distance of the two boxes and the NMS, results in a more precise bounding box.

3. ASFF was also used to learn the fused spatial weights dynamically.

4. This work is the first study utilizing the ELDDS1400C5 dataset.

5. This is the first research paper to utilize the state-of-the-art YOLOv8 for fault detection in PV solar cells.

The paper is organized as follows: Section II summarizes the related work and state of the art on photovoltaic solar cell defect detection, including those that use deep learning-based models and Section III describes the proposed model and underlying methodologies used to create it. Section IV presents the dataset, results, and discussion. Finally, Section V presents the study's pertinent conclusions and a list of future research directions.

II. RELATED WORK

Several studies have been conducted on automatic defect detection in solar cells, but only a few have applied EL imaging. Semantic segmentation can generate a pixel-level classification that allows object detection and classification of multiple objects. In [6], Pratt et al. developed the first semantic segmentation model that utilized the UNET architecture to detect and classify defects in PV modules. They trained and tested the proposed model on PV modules composed of monocrystalline and multi-crystalline silicon cells. The model performed better on cells made from mono-Si cells than on cells made from multi-Si cells. Tang et al. [18] developed a CNN-based model to classify four PV module defects: defect-free, micro-crack, break, and finger-interrupt. First, researchers used Generative Adversarial Networks (GANs) to generate a large dataset of high-resolution EL images from a small number of samples. The generated dataset was used to train four classifiers based on VGG16, ResNet50, Inception V3, and MobileNet. The experimental results showed that the proposed approach could accurately classify PV module defects in the EL images. El-Rashidy [1] developed a two-stage portable and efficient solar cell defect detection system in another study. The first stage employed K-means, MobileNetV2, and linear discriminant algorithms to cluster solar cell images and build detection models for each constructed cluster, followed by the development of a classifier model for the solar cell images of the formed clusters. The constructed linear discriminant classifier was used in the second stage to classify the cell image into the cluster label with the closest set of solar images. The defective solar cell was then identified using the model built using the assigned clusters. The proposed approach had the highest accuracy compared with other recent research frameworks. The authors of [5] developed an automatic system for PV defect classification in EL images using a light six-layer Convolutional Neural Network (CNN) architecture encompassing four convolutional layers and two fully connected layers. First, they performed various data augmentation operations to obtain additional data. Subsequently, they trained the model for 150 epochs with a batch size of 64. In addition to using minimal computational power and time, they achieved a high accuracy of 93.02%. Rodriguez et al. [19] presented a framework for segmentation and error detection of PV Modules. Firstly, they provided a list of all the existing solar panel EL image datasets, descriptions, and shortages. They introduced the ELDDS1400C5 custom dataset, which included 1,400 monocrystalline PV panel EL images with 109,200 cells. They trained an object detection model based on YOLOv5l for PV defect detection. Moreover, they performed a comparative analysis to evaluate the performance of the YOLOv5 model using three distinct loss functions, Cross Entropy (CE), Combo Loss (CL), and Focal Tervsky Loss (FTL). The best model achieved an mAP @0.5 of 0.778. Finally, they used image-gradient-based line localization to present a Cell segmentation mechanism for precise and reliable cell localization and to obtain superior PV panel representation.

In a recent study, Zhang and Yin [20] developed a system to detect three classes of defects in solar cell surfaces using an improved YOLOv5. First, deformable convolution was integrated into the CSP module. In addition, the feature extraction ability of the model was enhanced by incorporating the ECA-Net attention mechanism. Finally, they added tiny defect detection to boost the target detection accuracy at various scales. They also utilized MixUp and Mosaic data augmentation. The improved YOLOv5 achieved an mAP of 89.64%, which was 7.85% higher than that of the original version of YOLOv5.

In [21], a novel semantic transformer representation network (STRNet) was introduced for the purpose of achieving real-time crack segmentation at the pixel level in complex scenes. The STRNet is a carefully designed architecture incorporating several key elements to ensure efficient processing capabilities, including a squeeze and excitation attention-based encoder, a multi-head attention-based decoder, coarse upsampling, a focal-Tversky loss function, and a learnable swish activation function. Notably, the network exhibits high precision, recall, F1 score, and mIoU (mean intersection over union) performance, achieving values of 91.7%, 92.7%, 92.2%, and 92.6%, respectively. To verify the effectiveness of STRNet, the authors have compared its performance to that of other advanced networks, including Attention U-net, CrackSegNet, Deeplab V3+, FPHBN, and Unet++. Another study [22] reported on a high-performance deep-learning network for pixel-level segmentation of internal damages in concrete members using active thermography. Specifically, an attention-based generative adversarial

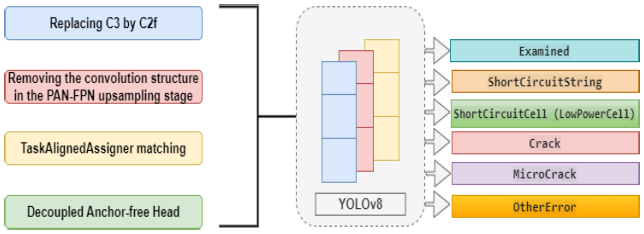


FIGURE 1. The proposed system architecture.

network (AGAN) was developed for training a novel internal damage segmentation network (IDSNet) that surpassed other state-of-the-art methods. Overall, the study showcases the potential of deep-learning approaches for efficiently and accurately detecting internal damages in concrete members using active thermography. Lewis et al. [23] developed a novel end-to-end dual encoder-decoder architecture, called Polyp Segmentation Network (PSNet), for pixel-level segmentation of polyps. The dual encoder and decoder in PSNet were created by combining several deep learning modules, including the PS encoder, transformer encoder, PS decoder, enhanced dilated transformer decoder, partial decoder, and merge module. In an extensive comparative study against 5 existing polyp datasets, PSNet outperformed state-of-the-art approaches in terms of mDice and mIoU, achieving 0.863 and 0.797, respectively. Overall, the results of this study demonstrated the effectiveness of the proposed PSNet architecture, which leverages multiple deep learning technologies to achieve superior segmentation performance.

The proposed model overcomes the shortcomings of models previously reported in the literature, such as achieving better performance using a less complex architecture and optimizing the state-of-the-art object detection model, YOLOv8. The architecture of the proposed system is illustrated in FIGURE 1.

III. METHODOLOGY

This section discusses the dataset, proposed models, and all technologies used to develop it.

A. THE DATASET

The dataset utilized in this study is the ELDDS1400C5 (ElectroLuminescence Defects DataSet) [19], which is a publicly available custom dataset of 1,400 monocrystalline PV panel EL images with 109,200 cells. It includes 21,000 bounding box-based annotations for the five defect types. Moreover, the dataset encompasses some minor errors such as dark spots and low-power cells. It contains five categories of defects, ShortCircuitString, ShortCircuitCell/LowPowerCell, Crack, MicroCrack, and OtherError (DarkSpot, Scratch, and Contamination).

The images in the dataset were acquired by Wuhan Solar Sunrise Photoelectric Technology Co., Ltd.'s Solar module EL Tester EL-1600/2400. The dataset contained 1187 training images and 190 testing images. FIGURE 2 depicts the class distribution of the defects in the ELDDS1400C5 dataset.

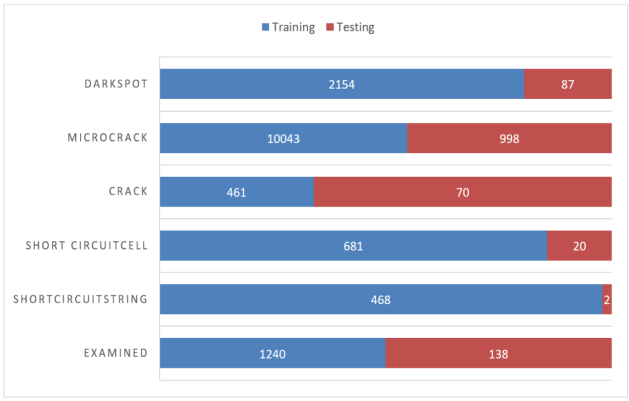


FIGURE 2. Class distribution of the ELDDS1400C5 dataset.

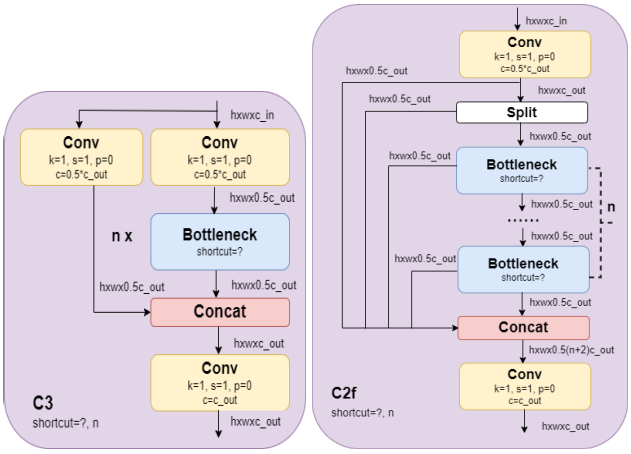


FIGURE 3. (a) C3 module, (b) C2f module.

B. DETAILS ABOUT THE TRAINING ENVIRONMENT

The Kaggle NVIDIA TESLA P100 GPU, CUDA11.1, Python3.8.8, and PyTorch1.11.0 were used. An Adam optimizer was used with a batch size of 32, initial learning rate of 0.01, 100 training epochs, and input image size of 640 × 640 pixels. The YOLOv8 model used yolov8s.pt while the original YOLOv5 model and the improved lightweight version used YOLOv5s.pt.

C. YOLOv8 MODEL

This study utilizes the most recent detector based on the YOLO architecture, YOLOv8 [24]. It is a single-stage anchor-free object detector. The key reasons for selecting this model were its flexibility, high accuracy, simplicity of training and deployment, and fast detection speed. It adds new features and enhancements to further boost the flexibility and performance. The main innovations encompass a new backbone network, anchor-free detection head, and new loss function that can run on CPUs or GPUs. Based on the scaling factor, YOLOv8 comes in different size models: nano (n), small(s), medium (m), large (l), and extra-large (x) scales to meet the needs of various scenarios. Unlike YOLOv5, the C2f structure with a richer gradient flow replaces the C3 structure of YOLOv5 in the backbone network and neck part,

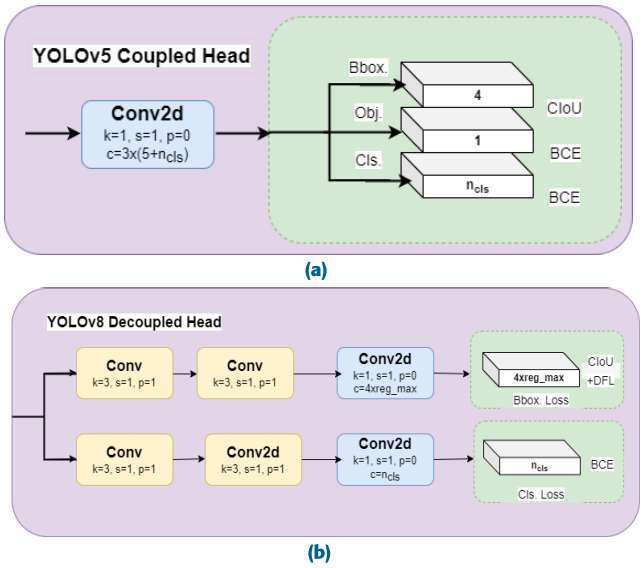


FIGURE 4. (a) YOLOv5 coupled head, (b) YOLOV8 decoupled head.

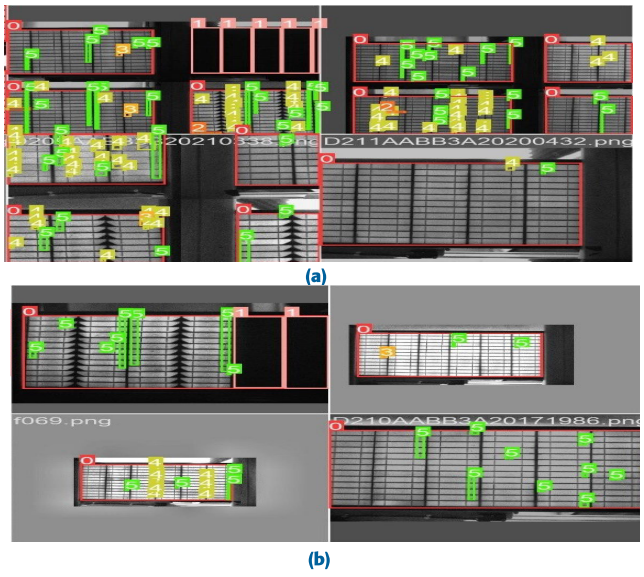


FIGURE 5. (a) Mosaic-augmented batch, (b) Mosaic-disabled batch.

and different channel numbers are modified for the various scale models. FIGURE 3 shows the key differences between C3 and C2f structures. As shown, there are more jump connections and Split operations. In addition, The Neck module's two convolutional connection layers were removed. Finally, the backbone's C2f block count ranges from 3-6-9-3 to 3-6-6-3. The decoupled classification and regression branches replace the previous objectness branch. The regression branch employs the integral form notation presented in Distribution Focal Loss (DFL).

The architecture of the head section in YOLO object detection system underwent significant changes in subsequent versions, YOLOv5 and beyond. The previous coupled head structure in YOLOv5 shared parameters between the classification and localization heads, as depicted in FIGURE 4 (a). This structure has since been replaced with a decoupled head architecture shown in FIGURE 4 (b). The new configuration effectively separates the detection and classification heads within the detection module, allowing for an Anchor-Based to Anchor-Free switch to be created. The resulting independent classification and localization heads could specialize and focus on their respective tasks, potentially leading to improved performance.

The YOLOv8 uses TaskAlignedAssigner positive sample allocation strategy instead of the previous IOU matching or unilateral ratio allocation methods for loss calculation, and it also introduces a Distribution Focal Loss. In addition, the Complete IoU (CIoU) loss was also utilized. Concerning data enhancement during training, the authors suggested turning off the mosaic data loader in the last ten epochs to boost the performance and achieve higher accuracy. Mosaic augmentation was accomplished by resizing and stitching together four images and selecting a random cutout, with the resulting mosaic image used to enhance object recognition capabilities of the model. This scheme allows for improved generalization, enabling recognition of objects in new locations or under partial occlusion as well as the ability to recognize objects on a smaller scale. Additionally, it reduces the need for a large mini-batch size during training. However, an important finding from related research is that applying this technique throughout the entire training phase reduces performance. As a result, mosaic augmentation is disabled for the last ten training epochs. FIGURE 5 illustrates the effects of mosaic augmentation on training batches, with one batch showing disabled mosaic augmentation for comparison to the mosaic-augmented batch.

D. YOLOv5 MODEL

Yolov5 is a well-known single-stage object detection model. It includes multiple basic models: small (s), medium (m), large (l), and extra-large (x). As the model width and depth increased, model complexity increased. The model has three main architectural sections: backbone, neck, and head. The backbone block is responsible for feature extraction. The neck network combines the gathered image features by the backbone model and then passes the integrated feature maps to the head, which creates predictions from the anchor boxes. Yolov5s is a lightweight version that can meet real-time requirements while consuming less computation; therefore, it was chosen as the baseline model.

E. INTEGRATION OF GLOBAL ATTENTION MECHANISM (GAM)

The attention mechanism is frequently used to extract essential information from input objects. The Global Attention Module (GAM) [25] is incorporated into the backbone and neck of the traditional YOLOv5s model for better object representation by ignoring unnecessary features while highlighting important features with a high correlation to the object to be detected. The entire process of the GAM module is shown in FIGURE 6.

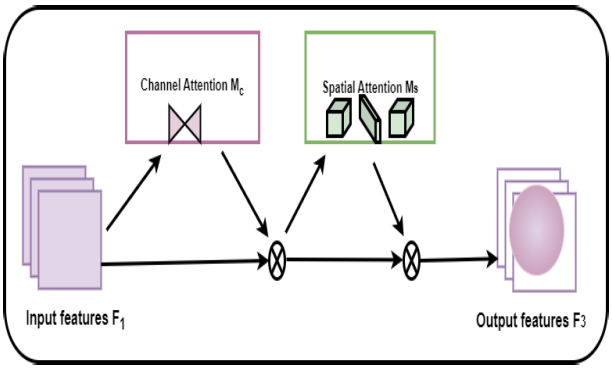


FIGURE 6. Schematic diagram of GAM attention process.

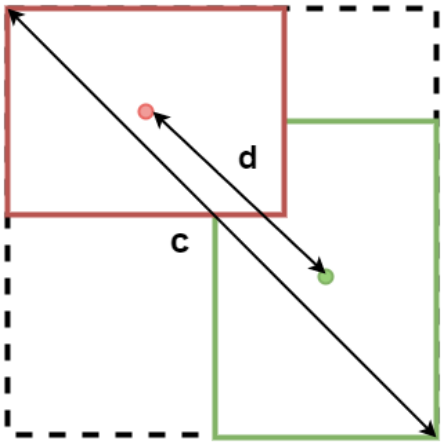


FIGURE 7. Details about the DIoU loss. The green, red, and black dashed boxes denote the predicted box, ground truth box, and C detection box, respectively.

GAM attention consists of two submodules: channel attention and spatial attention. If the input feature map is $F_1 \in \mathbb{R}^{C\times H\times W}$, the output F3 is given by Eq.1

$$F_3 = M_s(M_c(F_1) \otimes F_1) \otimes (M_c(F_1) \otimes F_1) \tag{1}$$

where M_s is the spatial attention map and M_c is the channel attention map, $\otimes$ represents element-wise multiplication. Although adding GAM increases the number of parameters in the network model, it allows the network to capture essential features more effectively. GAM improves the connection between channels and space and reduces the loss of information regarding defects in a PV module. Moreover, it highlights features of the global dimension.

F. INTEGRATION OF DIoU NMS LOSS

Object detection algorithms typically employ non-maximum suppression during prediction to discard redundant detection bounding boxes. The idea is to maintain the predicted bounding box with the highest score after passing through all the detection bounding boxes. Subsequently, the intersection ratio IoU between the remaining bounding boxes and the highest-score predicted bounding box was calculated. The detection bounding box was rejected if the IoU was lower than a certain threshold. Generally, the NMS method is effective. However, in the case of a dense target scene, the objects

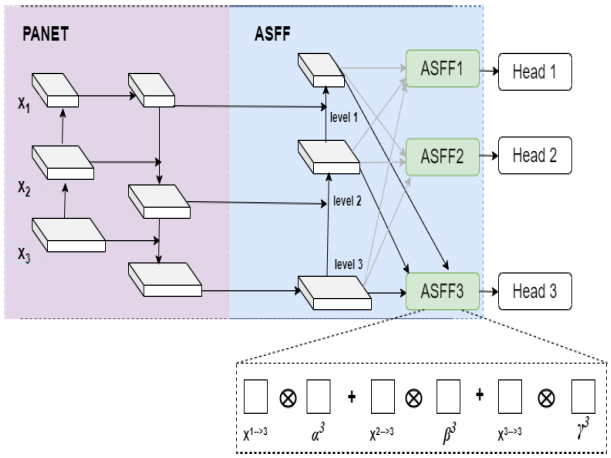


FIGURE 8. The head branch integrated with the ASFF module.

to be detected are too close to each other. Consequently, the overlapping area of the detection frames of the adjacent objects is too large, which consequently causes the NMS to reject the detection bounding boxes of a specific object incorrectly. The IoU loss (IoU) [26] is the most widely used localization loss in several object detection models. Regrettably, the network cannot be trained if the ground truth predicted boxes do not intersect because IoU = 0. IoU is given by Eq.2

$$IoU = \frac{|b \cap b^{gt}|}{|b \cup b^{gt}|} \tag{2}$$

As shown in FIGURE 7, in addition to the NMS, DIoU takes the overlap area and the center distance of the two boxes into account to accelerate convergence, making it more optimal and accurate for suppressing the predicted bounding box. Eq.3 and Eq.4 define DIoU and DIoU losses, respectively.

$$DIoU = IoU - \frac{\rho^2(b, b^{gt})}{c^2} \tag{3}$$

$$L_{DIoU} = 1 - DIoU = 1 - IoU - \frac{\rho^2(b, b^{gt})}{c^2} \tag{4}$$

where b and b^{gt} depict the centroids of the predicted and ground truth boxes, $\rho^2(b, b^{gt})$ or d represents the square of the Euclidean distance between their centers, and c^2 is the square of the diagonal length of the C detection box. The C detection box is the smallest rectangular box enclosing the predicted and ground truth boxes [27].

G. INTEGRATION OF ASFF DETECTION HEAD

In the prediction section, to improve the accuracy of the model and ensure that each space can dynamically merge different levels of feature information, Adaptive Feature space fusion (ASFF) is added to YOLOv5's original structure for feature fusion. ASFF is utilized in the head branch of YOLOv5 to optimize the spatial information of lower-level features and semantic information of upper-level features in the network. Adaptive weighting merges three feature images generated by the PAN module. FIGURE 8 shows the PANet structure of the proposed fused ASFF module.

TABLE 1. Evaluation metrics for all models.

Model	Precision	Recall	mAP@ 0.5	mAP@ 0.5-0.95
YOLOv5s _only	0.726	0.723	0.738	0.502
YOLOv5s+ GAM_only	0.748	0.744	0.754	0.511
YOLOv5s+ ASFF_only	0.773	0.712	0.751	0.499
YOLOv5s+ DIoU_only	0.741	0.719	0.744	0.5
YOLOv5s+ GAM+ASFF +DIoU	0.735	0.76	0.763	0.516

TABLE 2. Analysis of model complexity and TTA effect on mAP@0.5.

Model	mAP@0.5		FLOPs
	W/O TTA	With TTA	
YOLOv5s_only	0.739	0.766	16 G
YOLOv5s+ GAM_only	0.752	0.775	21.4 G
YOLOv5s+ ASFF_only	0.74991	0.766	25.8 G
YOLOv5s+ DIoU_only	0.74623	0.772	16 G
YOLOv5s+ GAM+ASFF+DIoU	0.76222	**0.777**	30.0 G
YOLOv8s	**0.765**	0.775	28.7 G

PANET outputs three feature layers: level 1, level 2, and level 3. The green box depicts how the ASFF features are combined. The features of levels 1, 2, and 3 are given as $X^{1\to3}$, $X^{2\to3}$, and $X^{3\to3}$respectively. Then,$X^{1\to3}$, $X^{2\to3}$, and $X^{3\to3}$ are multiplied by their corresponding weights and added together. Eventually, following the feature fusion process, the feature ASFF-3 is generated. Eq.5 shows the ASFF methodology

$$y_{ij}^{l} = \alpha^{3}.X_{ij}{}^{1\to l} + \beta^{3}.X_{ij}^{2\to l} + \gamma^{3}.X_{ij}^{3\to l} \tag{5}$$

where y_{ij}^{l}represents the new feature map, the weight parameters of the three feature maps are given by α^3, β^3, and γ^3, respectively. By contrast, The features of layers 1, 2, and 3 are denoted by $X_{ij}^{1\to l}$, $X_{ij}^{2\to l}$, and $X_{ij}^{3\to l}$respectively making it satisfy $\alpha^3 + \beta^3 + \gamma^3 = 1$ using the Softmax function and defining Eq. 6. In this, the numbers λ_{α}^{3}, λ_{β}^{3}, and λ_{γ}^{3} were used as control parameters.

$$\alpha^{3} = \frac{e^{\lambda_{\alpha}^{3}}}{e^{\lambda_{\alpha}^{3}} + e^{\lambda_{\beta}^{3}} + e^{\lambda_{\gamma}^{3}}} \tag{6}$$

The ASFF algorithm fully achieves the model's multiscale feature fusion by adjusting the feature fusion with the weight parameters.

IV. RESULTS AND DISCUSSION

In this section, the performance of the proposed model is tested and compared with state-of-the-art models using two different image datasets.

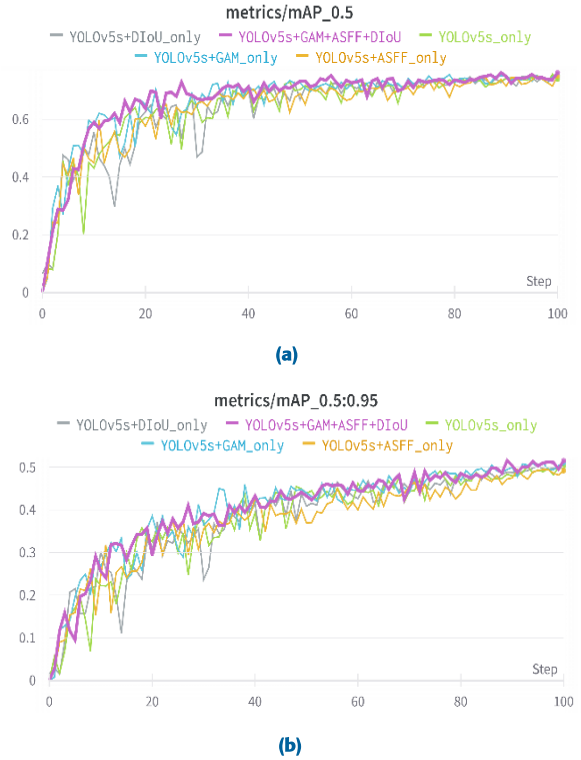


FIGURE 9. (a) mAP@0.5, (b) mAP@0.5-0.95.

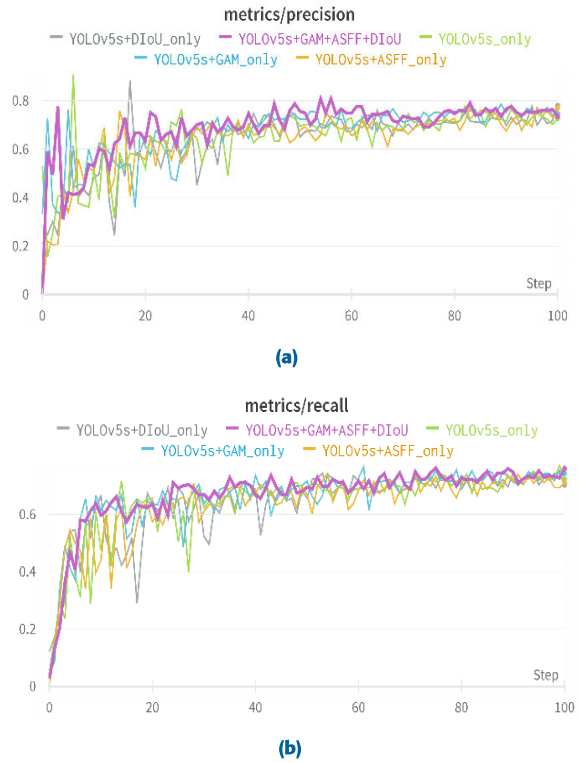
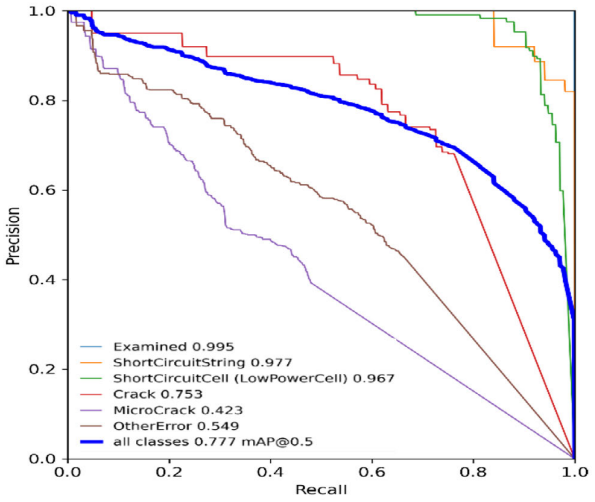


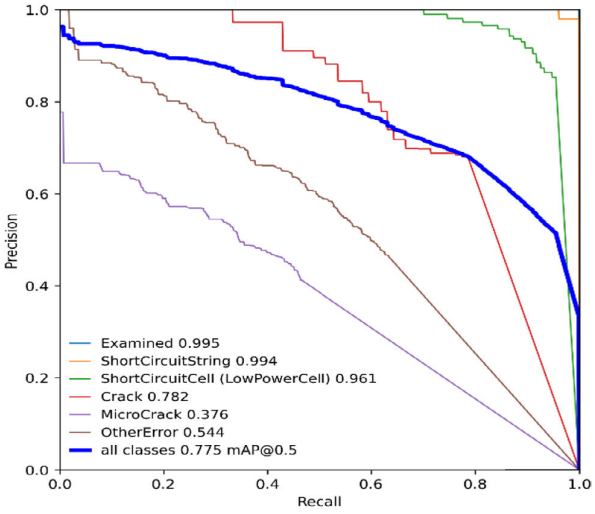
FIGURE 10. (a) Precision, (b) Recall.

A. EVALUATION METRICS

Precision (P), recall (R), average precision (AP), mean average precision (mAP), F1 score given by Eq.7, Eq.8, Eq.9,

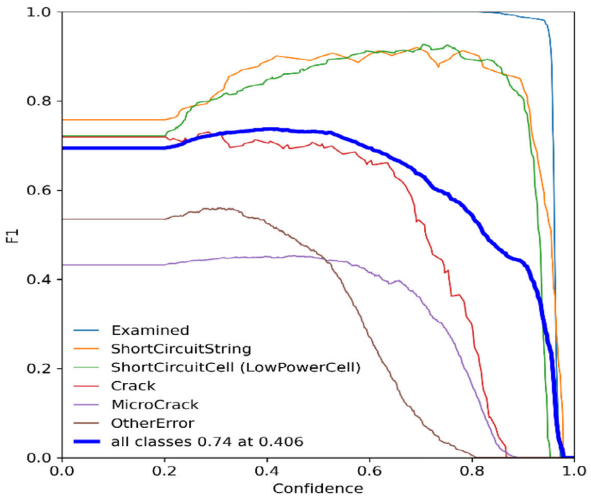


(a)

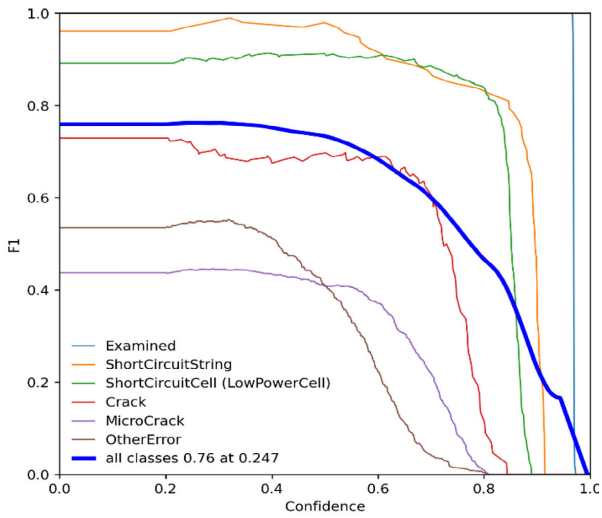


(b)

FIGURE 11. Precision-Recall curve (a) TTA-improved YOLOv5, (b) TTA-YOLOv8.



(a)



(b)

FIGURE 12. F1-curve (a) TTA-improved YOLOv5, (b) TTA-YOLOv8.

Eq.10 respectively, and Floating Point Operations (FLOPs) were used as performance evaluation metrics in this study. AP is the area under the PR curve, and mAP is the average AP for each class.

$$P = \frac{TP}{TP + FP} \tag{7}$$

$$R = \frac{TP}{TP + FN} \tag{8}$$

$$mAP = \frac{1}{n}\sum_{j=1}^{n} AP_j \tag{9}$$

where: $AP = \int_0^1 P(R)dR$, TP = True Positive, FP = False Positive, FN = False Negative

$$F_1 = \frac{2 \times (P \times R)}{(P + R)} \tag{10}$$

Precision measures the quality of a model's positive predictions. It is defined as the ratio of true positives to the total positive predictions. On the other hand, recall is also known as "sensitivity" or "true positive rate." This is defined as the ratio of positives that were correctly classified. Average precision represents the overall detection precision of the model. It is defined as the area enclosed by the Precision-Recall curve. The F1 metric assesses the trade-off between the precision and recall. It also serves as an indicator of the model's detection accuracy [28]. MAP is the arithmetic mean of the AP. It starts by computing the AP for each class and averaging them. The FLOPs are a measure of model complexity.

TABLE 1 lists the values of *Precision*, *Recall* *mAP*@0.5, *mAP*@0.50.95, respectively of the baseline and improved versions of the YOLOv5 model. As shown, integrating the baseline YOLOv5s with GAM, ASFF, and DIoU simultaneously outperformed the baseline model and other improved versions of YOLOv5. Figure 9 and 10 show the results in Table 1 for a deeper comprehension.

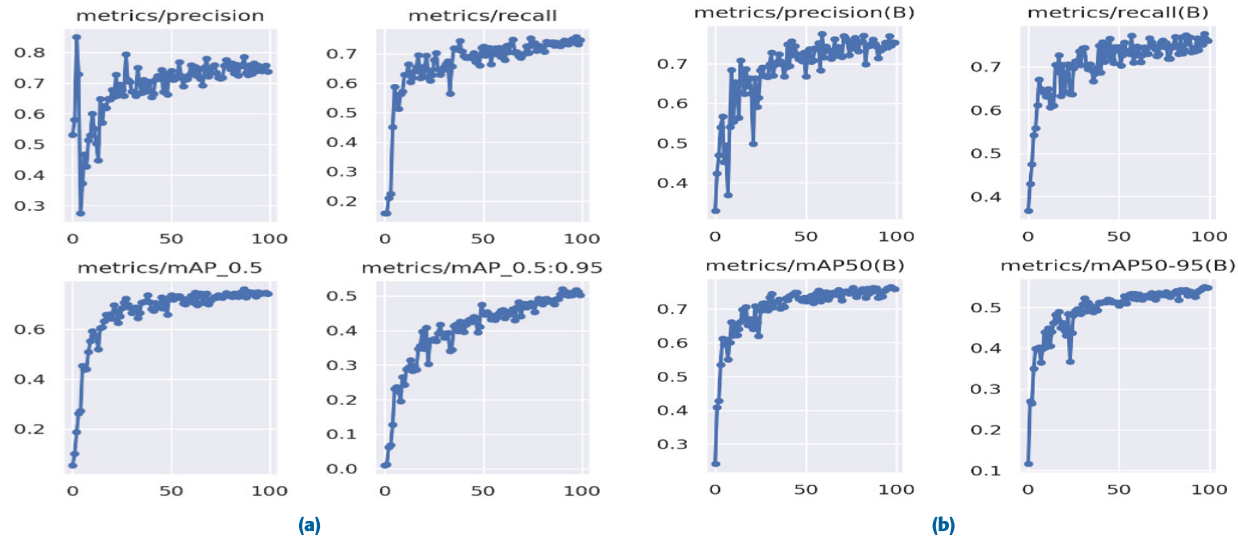


FIGURE 13. Evaluation metrics (a) TTA-improved YOLOv5, (b) TTA-YOLOv8.

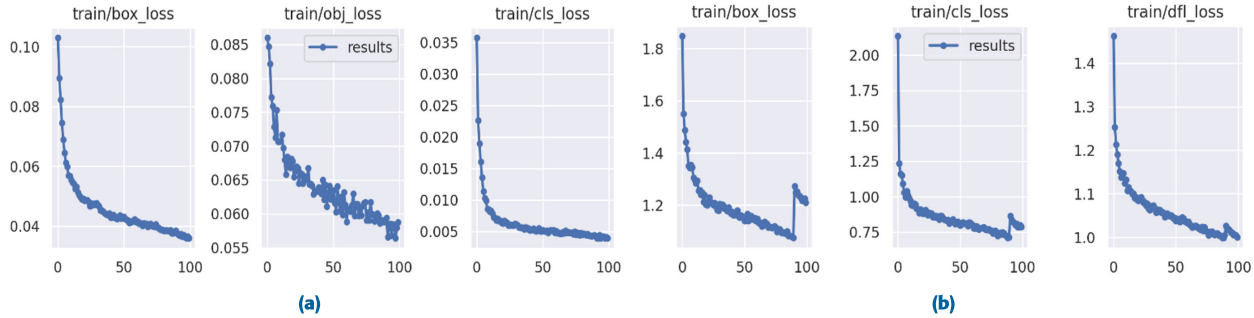


FIGURE 14. Loss curves (a) TTA-improved YOLOv5, (b) TTA-YOLOv8.

The model performance at inference time can be boosted by incorporating test-time augmentations (TTA) [29]. TTA means that a horizontal flip and three different resolutions are applied to each image, and the outputs of these augmentations were averaged to generate the final prediction. TABLE 2 summarizes the mAP@0.5 of the original YOLOv5s, improved YOLOv5 versions, and YOLOv8s with and without TTA. Using TTA, the mAP@0.5 of all models was improved. The proposed model outperformed the state-of-the-art YOLOv8 model in terms of mAP. Its mAP@0.5 has improved from 0.762 to 0.777, demonstrating the superiority of the proposed model. Regarding the inference speed, YOLOv8 surpassed all versions of the YOLOv5 model. While YOLOv8 requires approximately 6 ms to perform preprocessing, inference, and post-processing, the different versions of the YOLOv5 model require an average of 13 ms to perform the same operations as for image. The model complexity measured by the FLOPs indicates that the YOLOv8 model is less complex than the best-improved version of the YOLOv5 model, which is enhanced using GAM, ASFF, and DIoU simultaneously.

To further enhance the experimental results, the results of the best-proposed models were compared to those of a recent study conducted using the ELDDS1400C5 dataset and concluded in TABLE 3. The authors of [19] utilized the YOLOv5l model to detect six classes of defects. Using cross-entropy loss, the proposed YOLOv5s + GAM + ASFF + DIoU model outperformed not only their model in performance but also in model complexity.

TABLE 3. Comparison of the best-proposed models and recent studies.

model	mAP@0.5
YOLOv5l [19]	0.727
YOLOv5s+GAM+ASFF+DIoU+TTA[ours]	**0.777**
YOLOv8s+TTA [ours]	0.775

FIGURE 11 shows the Precision-Recall curves for the improved YOLOv5 and YOLOv8 models, respectively, after using the TTA. The precision-recall curve shows the trade-off between precision and recall for different thresholds. The larger the area under the curve, the higher is the recall and precision. For example, the TTA-improved YOLOv5 achieved an average mAP@0.5 of 0.777 for all classes, whereas the YOLOv8 achieved an average value of 0.775. Moreover, the TTA-Improved YOLOv5 performs better than the YOLOv8 model in detecting minor size defects such as Crack and MicroCrack, which validate the significant effect of the GAM attention module in detecting small targets by capturing features that are highly correlated to the target object.

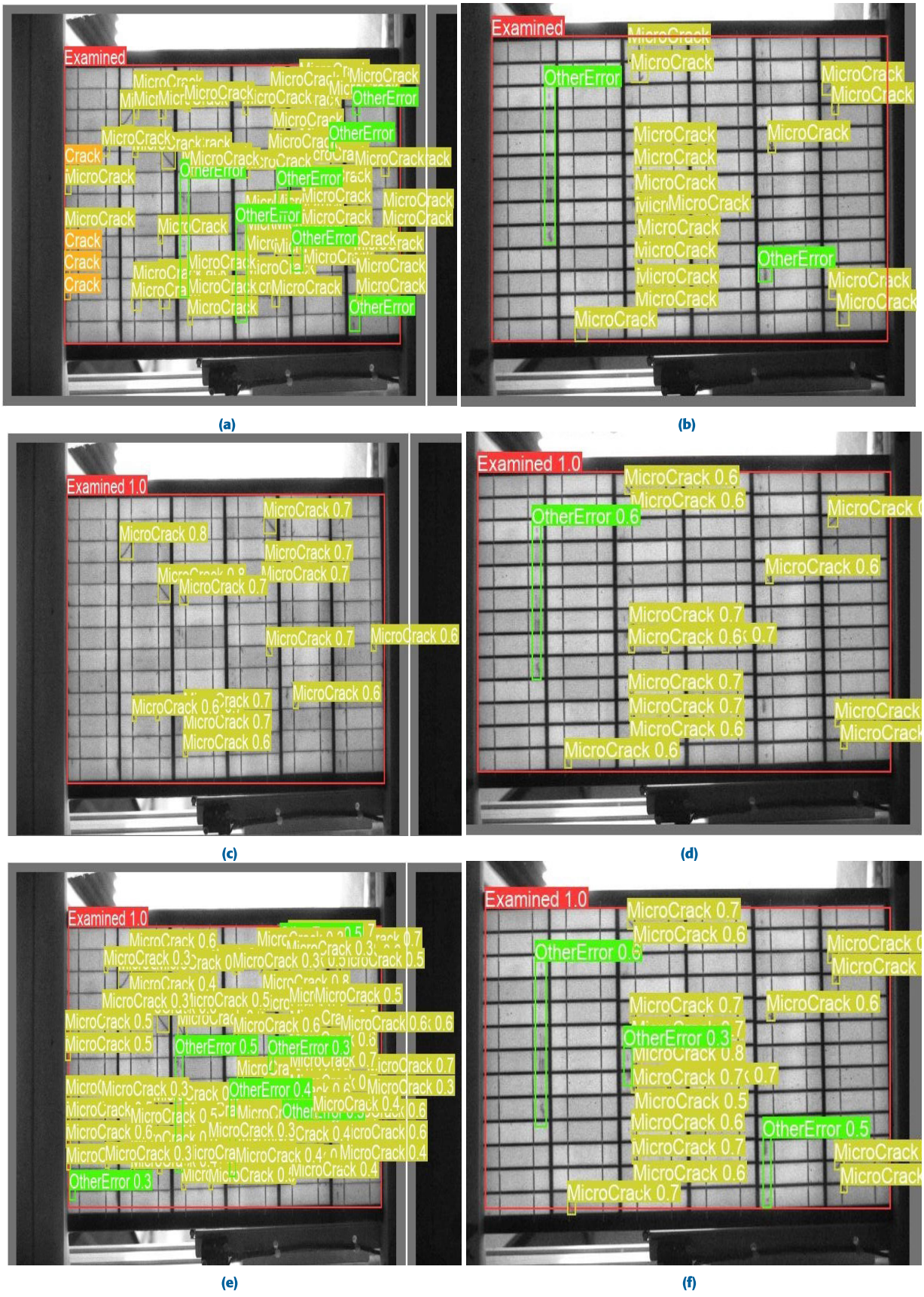
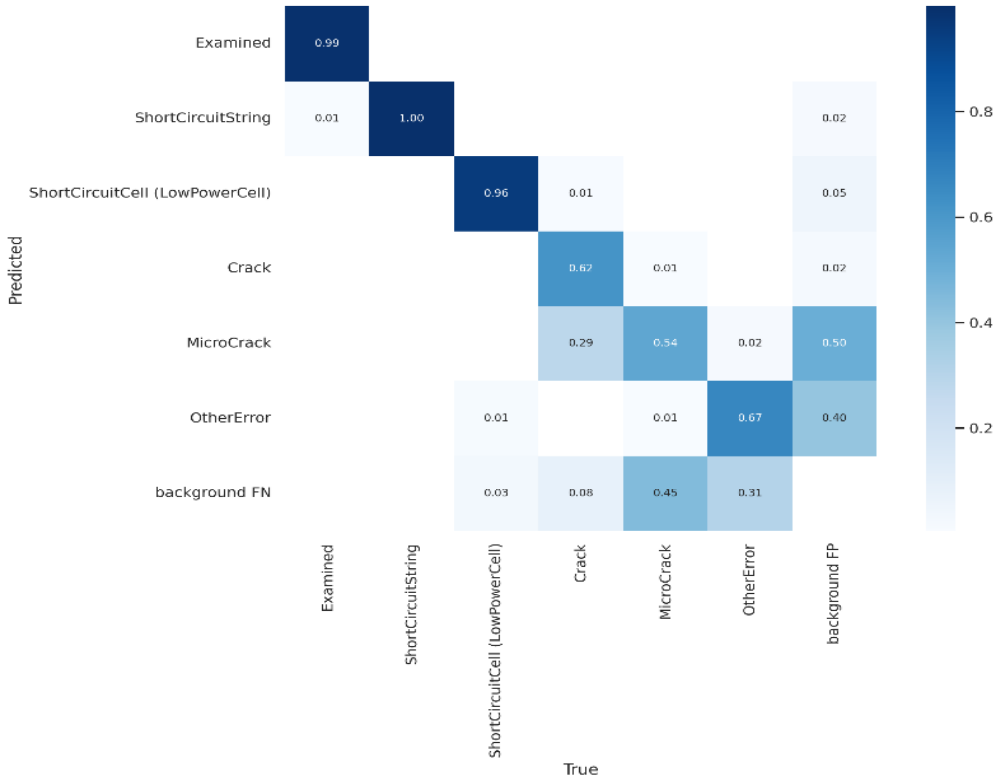
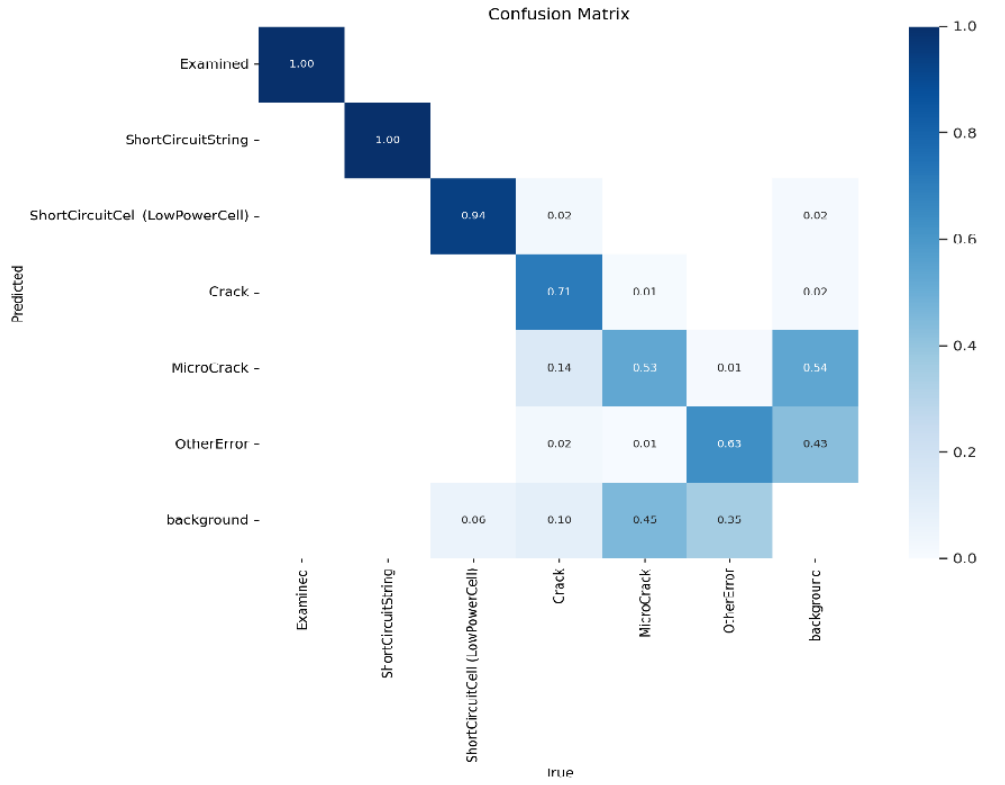


FIGURE 15. (a)&(b) Ground truth, (c)&(d) YOLOV8, (e)&(f) Improved YOLOv5.



(a)



(b)

FIGURE 16. Confusion matrix at conf = 0.2 and iou = 0.5 (a) TTA-improved YOLOv5, (b) TTA-YOLOv8.

FIGURE 12 depicts the F1 curve for the TTA-Improved YOLOv5 and TTA-YOLOv8 models. The horizontal axis represents the confidence level, whereas the vertical axis represents the F1 score, which lies in the interval [0, 1]. This assumes that precision and recall are equally important. The TTA-Improved YOLOv5 achieved an overall F1 score of 0.74 at a confidence level of 0.406. At this confidence level, optimum values of precision and recall can be obtained. On the other hand, TTA-YOLOv8 achieved an overall F1 score of 0.76 at a confidence level of 0.247.

The evaluation metrics and loss curves of the TTA-Improved YOLOv5 and TTA-YOLOv8 models during training are shown in FIGURE 13 and 14, respectively. In regards to object detection, precision, recall, and mAP curves are commonly used to evaluate model performance. Precision curves illustrate the percentage of precise bounding box predictions. Meanwhile, recall curves reveal the percentage of accurately predicted true bounding boxes. The classifier performance is thus appropriately captured by these metrics. It is important to note that higher precision and recall values indicate better model classification performance.

These curves indicate improved loss curve stability with increasing epochs. Before reaching epoch 20, the loss rapidly declined while recall, precision, and mAP all exhibited improvement. However, the reduction in loss function value and increases in mAP were subsequently slower. The object detection loss function comprising classification loss (cls_loss), bounding box location loss (box_loss), and confidence loss (obj_loss) was used to assess the models. The box_loss category assessed a model's capacity to accurately locate an object center point and whether the bounding box predictions enclosed detection targets. A smaller value of this loss indicates a more accurately predicted bounding box. The confidence loss is utilized to determine whether an object exists within the region of interest (predicted box). The smaller the value of the confidence loss, the greater the accuracy. Classification loss denotes the model's ability to correctly predict a given object class. The lower the loss value, the better is the classification.

FIGURE 15 depicts the predicted bounding boxes versus the predicted ones using the TTA-Improved YOLOv5 and TTA-YOLOv8 models.

FIGURE 16 displays the TTA-Improved YOLOv5 and TTA-YOLOv8 models' confusion matrices for classifying the six PV defects classes.

The actual classes are on the horizontal axis, whereas the predicted classes are on the vertical axis. The diagonal line in the middle of the confusion matrix indicates the significance of predicted results. The horizontal line represents false negatives, and the vertical line represents false positives. YOLO models can correctly classify most large-size defect classes such as Examined, ShortCircuitString, and ShortCircuitCell (LowPowerCell). However, some defects cause classification confusion owing to their inter-class similarity. Moreover, some errors are caused by classifying an object as background or background as an object.

V. CONCLUSION

The present paper posits a novel framework for the application of deep learning towards the automatic detection and classification of defective PV modules in the context of EL imaging. In summary, this paper presents a comparative analysis between the state-of-the-art YOLOv8 and an Improved YOLOv5s model for developing a computer vision-based PV defect detection system. Compared with the baseline YOLOv5s, the improved version achieved a good balance of performance and efficiency for detecting faults in PV modules. Improved-YOLOv5 starts with YOLOv5s as the baseline and then adds the GAM module to the network neck and backbone to improve the network's feature extraction. Furthermore, the ASFF was incorporated into the Head branch to maximize the network's upper-level semantic information and lower-level feature spatial information. Finally, the default NMS loss function was replaced by DIoU. The employment of EL imagery permits accurate discrimination between faulty and functional solar cells. Thus, the proposed framework is trained and assessed on the ELDDS1400C5 dataset, which comprises 2,426 high-resolution solar cells images. Additionally, this study elucidates the significant contribution of TTA in elevating the performance of deep learning models during inference for the aforementioned task. The multiple-scale defects in PV cell EL images are detected efficiently under complex background conditions using the proposed deep learning-based approach. The experimental results demonstrated that the proposed model, which utilizes EL imagery, outperformed the YOLOv8 model in both detection and classification of six different defects in PV solar cells. The proposed model achieved a mAP of 76.3% at an IoU threshold of 0.5, representing a 2.5% increase in performance compared to the original YOLOv5 model. This study suggests that the proposed model shows great promise for applications in automated defect detection and quality control in the PV industry. Future directions will include deploying the model as a mobile application to serve as a real-time monitoring system for defect detection in PV modules. Moreover, the interpretation and detection of defects in EL images will be used to predict the I-V curve. It will focus on establishing a relationship between the panel's electrical performance and the visual quality of its cells using only EL images.This research has important implications for the photovoltaic industry, as it offers a non-invasive and efficient method for automated quality control that can enhance the reliability and lifespan of PV modules.

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